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The bridge between two worlds: psychoanalysis and fMRI

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Abstract: In recent years, a connection between psychoanalysis and neuroscience has been sought. The meeting point between these two branches is represented by neuropsychanalysis. The goal of the relationship between psychoanalysis and neuroscience is to test psychoanalytic hypotheses in the human brain, using a scientific method. A literature search was conducted on May 2015. PubMed and Scopus databases were used to find studies for the

inclusion in the systematic review. Common results of the studies investigated are represented by a reduction, a modulation, or a normalization of the activation patterns found after the psychoanalytic therapy. New findings in the possible and useful relationship between psychoanalysis and neuroscience could change the modalities of relating to patients for psychoanalysts and the way in which neuroscientists plan their research. Researchers should keep in mind that in any scientific research that has to do with people, neuroscience and a scientific method cannot avoid subjective interpretation.

Keywords: fMRI; neuropsychanalysis; neuroscience; psychoanalysis; psychodynamic psychotherapy.

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Introduction

History

In recent years, a growing relationship between psychoanalysis and neuroscience has been sought (Shevrin et al., 1992; Kandel, 1999; Beutel et al., 2003; Northoff, 2007). The meeting point between these two branches is represented by neuropsychanalysis (Solms and Turnbull, 2011). Discussing neuropsychanalysis means speaking about Freud's attempts to translate psychological concepts into the language of neuroanatomy and neurophysiology (Freud, 1950). Freud was specifically interested in revealing the psychological predispositions of psychodynamic processes. However, he did not focus so much on the psychological functions during that time, as much as he did on the primary focus of neuroscience and cognitive neuroscience (Solms and Saling, 1986). Owing to the insufficient number of diagnostic neuroscientific tools required to link neuronal mechanisms and psychodynamic concepts, Freud had to abandon his neuropsychanalytic attempts (Northoff, 2012).

Today, our understanding of normal and abnormal mental functions comes through the development of neuroscience (Etkin et al., 2005). In recent decades,

modernization of assessment tools and diagnostic studies of the brain have given the chance to clarify the connection between the mind and the brain (Dobbs, 2007; Lichterman, 2010). Electroencephalography (EEG), introduced around the 1930s (Berger, 1929), represented the beginning of a capability to measure and observe the dynamic aspects of the brain activity under changing functional conditions (Moruzzi and Magoun, 1949). The development of Magnetoencephalography (MEG), introduced by Cohen (Cohen, 1968), represents a further substantial advance, allowing us to study the neural dynamics associated with mental events at the millisecond level, with increasing anatomical precision. However, the introduction of functional neuroimaging tools (functional magnetic resonance imaging, fMRI; position emission tomography, PET; and single-photon emission computed tomography, SPECT), from the 1990s, have allowed us to observe the neurodynamic processes under changing psychological conditions (Solms and Turnbull, 2011).

Backgrounds

Difficulties in creating a connection between psychoanalysis and neuroscience were evidenced because different approaches in the study of the mind have been used. Each approach develops a different representation of reality. However, if we assume that both approaches describe the mind in its more complex functionality, we can conclude that their concepts have common points (Ouss-Ryngaert and Golse, 2010; Kessler et al., 2013). New findings in the possible useful relationship between psychoanalysis and neuroscience could change the ways of relating to patients for psychoanalysts and the ways in which neuroscientists plan their research (Ouss-Ryngaert and Golse, 2010; Tordjman, 2010; Rizzolatti et al., 2014).

Mechelli (2010) considered three possibilities to integrate psychoanalysis and neuroscience. The first possibility is that mapping psychoanalytically defined phenomena in the brain could provide biological validity to these phenomena (Kandel, 1999; Solms and Turnbull, 2002). The second possibility is represented by the fact that brain science findings could be used to test psychoanalytic theories (Rubinstein, 1976; Hobson and McCarley, 1977; Kandel, 1998; Kandel, 1999; Solms and Turnbull, 2002; Green, 2003; Schore, 2003). The last possibility is represented by the theory that the neuroscientific method could be used to measure psychotherapy progresses and outcomes (Kandel, 1998; Kandel, 1999; Solms and Turnbull, 2002; Green, 2003).

Kessler's opinion to bridge both sides is represented by the system of operationalized psychodynamic

diagnosis (OPD) (OPD-Task-Force, 2008). The OPD System is a 1- to 2-h psychodynamic interview based on four psychodynamically relevant diagnostic axes, associated to the fifth 'International Classification of Mental and Behavioral Disorders' (ICD-10) axes (WHO, 1992). With this procedure, Kessler suggests reaching an intermediate level, removing as many contradictions as possible and preserving as many dynamic contents as possible (Kessler et al., 2013). Other methods to evaluate subjective experiences, as objective as possible, are represented by the Maladaptive Interpersonal Patterns Q-Sort (MIPQS) (Zimmermann et al. 2014), the Interpersonal Relations Picture Set (IRPS) proposed by Boeker et al. (2013), the Structured Interview for Personality Organization (STIPO) (Clarkin et al., 2004), and the Karolinska Psychodynamic Profile (KAPP) (Weinryb et al., 1991a,b).

Neuropsychanalysis aims to link psychodynamic concepts to specific psychological functions localized in particular brain regions (Schore, 1997; Northoff, 2012). In this context, many efforts have been made regarding dreams (Solms, 1997; Solms, 2000; Solms and Turnbull, 2002; Mancia, 2004; Hobson, 2009; Northoff, 2011), the ego (Northoff et al., 2007; Carhart-Harris and Friston, 2010), defense mechanisms (Fonagy, 2003; Northoff and Boeker, 2006; Northoff, 2007; Feinberg, 2011; Schmeing et al., 2013), empathy (Gallese et al., 2007), neuronal mechanisms underlying drives (Solms, 1996; Panksepp, 1998; Fonagy, 2003) the unconscious and its relation to memories (Kandel, 1998; Mancia, 2004, 2006a,b; Peres et al., 2008), and primary and secondary processes (Carhart-Harris and Friston, 2010).

The development of neuropsychanalysis began from the need to give scientific proof on the clinical effectiveness of psychoanalysis (Panksepp and Solms, 2012). In fact, while many brain imaging studies reported neural effects in patients treated with cognitive behavioral therapy and interpersonal therapy (for recent reviews, see Roffman et al., 2005; Kumari, 2006; Linden, 2006; Beauregard, 2007; Frewen et al., 2008), until 15 years ago, no studies were performed to demonstrate similar effects on psychodynamic and psychoanalytic therapies.

There are important and numerous variables to control when researchers conduct a neuroimaging study that involves psychoanalytic or psychodynamic treatments. The main methodological problems are represented by design and translation problems (for a review, see Boeker et al., 2013). The design problems address issues about the psychotherapist (personality, empathy, attachment style), the patient, and the therapeutic relationship. Translation problems deal with different levels: personal and neuronal, generality vs. individuality, content vs. organization, and psychodynamic vs. process levels.

Recently some authors reported neurobiological and (in some cases also) clinical changes in different kinds of subjects treated with psychodynamic psychotherapy (Overbeck et al., 2004; Saarinen et al., 2005; Lai et al., 2007; Lehto et al., 2008; Beutel et al., 2010; Kessler et al., 2011; Buchheim et al., 2012; de Greck et al., 2013; Gawrysiak et al., 2013; Kehyayan et al., 2013; Taubner et al., 2013).

Moreover, several studies demonstrated the effectiveness of short-term and long-term psychoanalytic treatments in randomized controlled trials (Gabbard et al., 2002; Leichsenring et al., 2004; Leichsenring and Rabung, 2008; Leichsenring and Rabung, 2011). At the contrary, some authors thought that the methodology of randomized controlled trials is not applicable to long-term psychotherapy lasting several years or to psychoanalytic therapy (Seligman, 1995; Wallerstein, 1999).

Purpose of the systematic review

Until now, a small number of studies comparing fMRI scans before and after psychoanalytic psychotherapy have been conducted. The demonstration of changes in the brain, secondary to psychoanalytic treatment, can provide valuable information about the workings of the mind. In our opinion, the demonstration mentioned before must not be used to give validity to the psychoanalytic method, already demonstrated by a century of clinical use and some empirical data (for a review, see Leichsenring, 2005). The goal of the relationship between psychoanalysis and neuroscience is to test psychoanalytic hypotheses in the human brain, using a scientific method. Researchers should keep in mind that in any scientific research that has to do with people, neuroscience and a scientific method cannot avoid subjective interpretation.

Materials and methods

A literature search was conducted on May 2015. PubMed and Scopus databases were used to find studies for the inclusion in the systematic review. The keywords used for the literature search were: 'psychoanalysis', 'psychoanalytic psychotherapy', 'psychodynamic', 'psychodynamic psychotherapy', 'fMRI', 'functional neuroimaging', 'functional magnetic resonance', and 'functional magnetic resonance imaging'. In each search, the keywords were used together with the logical operator 'and'. Keywords used in the literature selection criteria are summarized in Table 1.

The total number of articles retrieved by using PubMed and Scopus databases were 86 ($n=86$). Sixty-eight ($n=68$) abstracts were excluded because they were review articles or articles not written in English, or they did not speak about correlations between

psychoanalytic or psychodynamic treatments and fMRI findings. For these reasons, 18 ($n=18$) full papers were examined. Eleven ($n=11$) studies were excluded because the authors conducted an fMRI scan only after the psychoanalytic or the psychodynamic interventions. Only the studies comparing fMRI results before and after the psychoanalytic or the psychodynamic therapies in adult subjects were taken into account. In this way, seven ($n=7$) potentially relevant studies were obtained for the systematic review (Figure 1).

Discussion

There are seven studies (Beutel et al., 2010; Buchheim et al., 2012, 2013; de Greck et al., 2013; Fischmann et al., 2013; Gawrysiak et al., 2013; Wiswede et al., 2014) in past research comparing results before and after a psychotherapeutic treatment in adults. While five studies used a psychodynamic treatment, only two studies (Buchheim et al., 2013; Fischmann et al., 2013) used a psychoanalytic therapy (Table 2). There are some similarities and differences between psychoanalysis and dynamic psychotherapy (for a review, see Rangel, 1954).

In both studies using a psychoanalytic treatment, the number of participants was too small even though there

Table 1: Keywords and logical operators used in the literature search.

Psychoanalysis	and	fMRI
Psychoanalytic psychotherapy		Functional neuroimaging
Psychodynamic psychotherapy		Functional magnetic resonance
		Functional magnetic resonance imaging

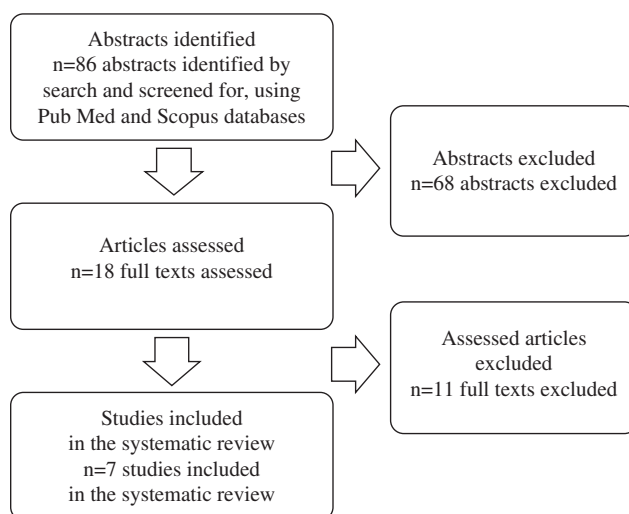


Figure 1: Flow chart of the systematic review.

Table 2: Studies included in the systematic review.

Authors	Scanner	Participants	Type of therapy	Methods in selecting stimuli	Current psychotropic medications	Recorded variables	Main findings
Buchheim et al., 2013	3.0 Tesla	A female dysthymic patient with narcissistic traits	Standard long-term psychoanalysis with a frequency of two face-to-face sessions per week	Adult Attachment Projective Picture System	Unknown	– The Psychotherapy Process Q-Set (Jones, 2000) – Collegium Internationale Psychiatrica Scalarum-depressiveness scale (Zerssen, 1976) – The outcome questionnaire-burden subscale (Haug et al., 2004)	The results show the therapy hours modulated the activation of the ventrolateral and the dorsolateral prefrontal cortex, the perigenual portion of the medial prefrontal cortex, the posterior cingulate and precuneus, the middle temporal gyrus, and the anterior tip of the inferior temporal gyrus, and the occipital/calcarine cortex
Fischmann et al., 2013	3.0 Tesla	16 Patients with recurrent major depressive disorders and 18 healthy volunteers	3-Year psychoanalytic treatment	Operationalized psychodynamic diagnosis	Patients were completely free of psychotropic medication at the time of study	– Dream changes analyzed by the method of Moser and von Zeppelin (1996) – Beck Depression Inventory – Symptom Check List 90 – Revised Version	After treatment, it was found that there was a marked reduction in activation pattern occipital, left and right hippocampus, left and right thalamus, left precuneus, left middle cingulate cortex, left inferior frontal gyrus/pars orbitalis, left precentral gyrus, vermis, left inferior frontal gyrus, right middle temporal gyrus, superior parietal lobule, cerebellum, and right and left putamen. Depressive symptoms improved significantly
Beutel et al., 2010	1.5 Tesla	9 Patients with panic disorder and 18 healthy controls	4 Weeks short-term psychodynamic treatment	Emotional linguistic go/nogo task	Five patients were free of psychotropic medication, and four continued their previous medication with antidepressants (2 SSRI; 2 tri-/tetracyclic antidepressants).	– The Body Sensations Questionnaire – Agoraphobic Cognitions Questionnaire (ACQ) – State-Trait Anxiety Inventory	Fronto-limbic activation patterns were normalized. New activation areas (caudate nucleus) were found in patients At the end of treatments, panic-related symptoms and bodily sensations improved significantly
Buchheim et al., 2012	3.0 Tesla	16 Recurrently depressed unmedicated outpatients and 17 controls	15 Months long-term psychodynamic psychotherapy	Adult Attachment Projective Picture System	Patients were completely free of psychotropic medication at the time of study	– Beck Depression Inventory – Global Severity Index from the revised Symptom Check List	The activation in the left anterior hippocampus/amygdala, subgenual cingulate, and medial prefrontal cortex found before treatment, were reduced after 15 months of long-term psychodynamic therapy. Clinical improvements were reported

Table 2 (continued)

Authors	Scanner	Participants	Type of therapy	Methods in selecting stimuli	Current psychotropic medications	Recorded variables	Main findings
de Greck et al., 2013	1.5 Tesla	15 Somatoform disorder patients and 15 healthy control subjects	Multimodal psychodynamic psychotherapy	'Japanese and Caucasian Facial Expressions of Emotion' battery and the 'Japanese and Caucasian Neutral Faces' battery	During the first fMRI session, five patients were treated with duloxetine. During the second fMRI session, one patient continued duloxetine treatment, and one other patient continued the treatment with duloxetine and trimipramine	– Toronto Alexithymia Scale-20 – Symptom Check List 90 – Revised Version, somatization sub-scale – Beck Depression Inventory – Tübingen Affekt Batterie	Increase in hemodynamic responses in the bilateral parahippocampal gyrus and in the left inferior temporal gyrus. Decrease in hemodynamic responses in the left putamen. As result, the brain activity in the bilateral parahippocampal gyrus was normalized. Significant reduction of somatization, alexithymia, and depressive symptoms
Gawrysiak et al., 2013	1.5 Tesla	A woman with major depression and breast cancer	Eight 1-h sessions of pragmatic psychodynamic psychotherapy	Alternating passages of preferred and neutral music	Unknown	– Beck Depression Inventory-II – The Hamilton Rating Scale for Depression – Beck Anxiety Inventory – Environmental Reward Observation Scale – Multidimensional Scale of Perceived Social Support	After eight 1-h session of therapy, blood oxygenation level dependence signal in the ventral-medial prefrontal cortex and in the left orbital prefrontal cortex were higher, demonstrating a return to a healthy condition. Depression scores were improved
Wiswede et al., 2014	3.0 Tesla	18 Unmedicated patients with recurrent major depressive disorder and 17 healthy subjects	8 Months of psychodynamic therapy	Operationalized psychodynamic diagnosis	Patients were completely free of psychotropic medication at the time of study	– Beck Depression Inventory	After 8 months of psychodynamic therapy, the differences in brain activity between patients and controls were disappeared. Depression scores were improved

were big differences. In fact, while Buchheim et al. (2013) assessed only one female dysthymic patient with narcissistic traits, Fischmann et al. (2013) investigated 16 patients with recurrent major depressive disorders. Moreover, only Fischmann et al. compared patients to healthy volunteers.

The studies used different psychotherapy durations, different methods to extract individualized and personally relevant stimuli, and a different number of fMRI scans. Buchheim et al. (2013) assessed the patient at monthly intervals for 15 years. Each month, immediately after therapy sessions, the patient underwent a fMRI scan. During the scan, the patient looked at pictures divided in two groups: 'neutral' and 'personally relevant'. The personally relevant pictures evoked the core conflict in the patient.

As the Fischmann et al. (2013) study is still ongoing, the author presented only the results of a single case. A female patient underwent a 3-year psychoanalytic treatment during which she was tested three times in the fMRI scans. Dreams and OPD interviews were conducted to extract individualized and personally relevant stimuli relating to depressive symptoms.

The common result from the studies investigated in this systematic review is represented by a reduction, a modulation, or a normalization of the activation patterns found before and after the psychoanalytic psychotherapy. Some differences between the studies can be highlighted. In the Buchheim et al. (2013) study, at the fMRI scan, the pictures described as 'personally relevant' resulted in activation of the ventrolateral and the dorsolateral prefrontal cortex, the perigenual portion of the medial prefrontal cortex, the posterior cingulate and precuneus, the middle temporal gyrus, and the anterior tip of the inferior temporal gyrus, and the occipital/calcarine cortex. In particular, the results showed as the quality of the therapy sessions that had immediately preceded the scan modulated the posterior cingulate cortex. The Fischmann et al. (2013) study was composed by two experiments: a dream-word experiment and an OPD experiment. In the dream-word experiment, the activation pattern in T1 (beginning of therapy) concerned the left inferior frontal, the left superior frontal, the left intraparietal cortex (angular gyrus, middle temporal gyrus), the left middle occipital gyrus, the right inferior frontal gyrus pars orbitalis, the right temporal pole, the left precuneus, the left anterior cingulate cortex, the left superior medial gyrus, the right inferior frontal gyrus/pars orbitalis, the right middle temporal gyrus, the left superior medial gyrus, and the left medial frontal gyrus. The pattern decreased from T1 to T2 (after 7 months of therapy), and the disappearance in T3 (after 15 months of therapy) assumes that the special

importance of the dream content had been lost and had been experienced by the subject in the same way as the neutral story. Similarly in the OPD experiment, the comparison between OPD sentences and relaxation showed an activation pattern at the T1 occipital, the left and the right hippocampus, the left and the right thalamus, the left precuneus, the left middle cingulate cortex, the left inferior frontal gyrus/pars orbitalis, the left precentral gyrus, vermis, the left inferior frontal gyrus, the right middle temporal gyrus, the superior parietal lobule, the cerebellum, and the right and the left putamen. A marked reduction from T1 to T2 and T3 was found.

Five studies (Beutel et al., 2010; Buchheim et al., 2012; de Greck et al., 2013; Gawrysiak et al., 2013; Wiswede et al., 2014) investigated pre- to post-fMRI changes in patients treated with psychodynamic therapy. All studies compared patients to control subjects with the exception of one (Gawrysiak et al., 2013). Patients enrolled in these studies had different diagnoses that involve different pretreatment brain activation areas. However, these data are not so relevant for the aim of this systematic review. In fact, what should be stressed is the evidence that equal studies that used psychoanalytic treatment and patients treated with psychodynamic therapy showed a normalization of pre- to post-brain activation areas.

After treatment, recurrent unmedicated depressed patients showed a reduction in the activation in the left anterior hippocampus and amygdala and medial prefrontal cortex (Buchheim et al., 2012) or a disappearance of the hyperactivity in the limbic and subcortical regions, including the amygdala and basal ganglia (Wiswede et al., 2014). These results supported the models of neurobiological effects of psychotherapy in depressed patients (Mayberg, 1997; Mayberg, 2003; DeRubeis et al. 2008). Moreover, they confirmed previous findings of a normalization of limbic hyperactivity after cognitive behavioral psychotherapy (Fu et al., 2008).

Somatoform disorder patients showed, after psychotherapy, an increase in hemodynamic responses in the bilateral parahippocampal gyrus and in the left inferior temporal gyrus and a decrease of hemodynamic responses in the left putamen. As a result, the brain activity in the bilateral parahippocampal gyrus was normalized (de Greck et al., 2013).

In patients with panic disorder Beutel et al. (2010) found that 4 weeks short-term psychodynamic therapy normalized the fronto-limbic activation patterns. On the other hand, new activation areas (caudate nucleus) were found only in patients. The authors stressed the fact that normalization results were not dissimilar to previous findings on cognitive-behavioral treatments.

Gawrysiak et al. (2013) used fMRI scans as a method to assess treatment outcomes of the pragmatic psychodynamic psychotherapy in a depressed patient. Neuroimaging studies in depressed patients reported alterations in brain regions involved in pleasure and reward processing (Keedwell et al., 2005; Epstein et al., 2006; Schaefer et al., 2006; Osuch et al., 2009). After eight, 1-h sessions of therapy, the blood oxygenation level dependence (BOLD) signal in the ventral-medial prefrontal cortex and in the left orbital prefrontal cortex was higher, demonstrating a return to a healthy condition.

Moreover, all studies reported a clinical improvement after the psychodynamic therapy. The normalization of the activation areas in depressed patients dealt with reductions in the Beck Depression Inventory (BDI) (Hautzinger et al., 1992; Beck et al., 1996a; Buchheim et al., 2012; Wiswede et al., 2014) and the Global Severity Index (GSI) from the revised Symptom Check List (SCL-90-R) (Derogatis, 1993; Franke, 1995; Buchheim et al., 2012).

de Greck et al. (2013) reported a significant reduction of somatization (Symptom Check List 90 – Revised Version, SCL-90 somatization sub-scale, German version), alexithymia (Toronto Alexithymia Scale-20, TAS-20) (Bagby et al., 1994; Bressi et al., 1996), depressive symptoms (BDI), and a significant reduction in error rates in an emotion recognition test ('Tübinger Affekt Batterie' TAB, German version of 'Florida Affect Battery' FAB) (Bowers et al., 1999).

At the end of treatments, panic-related symptoms, such as anxiety (STAI) (Laux et al. 1981), frequency of panic-related cognitions (assessed with the Agoraphobic Cognitions Questionnaire, ACQ), and bodily sensations (assessed with The Body Sensations Questionnaire, BSQ) (Chambless et al., 1984) improved significantly (Beutel et al., 2010).

Gawrysiak et al. (2013) reported improvements in Beck Depression Inventory-II (BDI-II) (Beck et al., 1996b), The Hamilton Rating Scale for Depression (HRSD) (Hamilton, 1960), Beck Anxiety Inventory (BAI) (Beck and Steer, 1993), the Environmental Reward Observation Scale (EROS) (Armento and Hopko, 2007), and the Multidimensional Scale of Perceived Social Support (MSPSS) (Zimet et al., 1988).

Different types of psychodynamic therapy were used to treat patients: short- and long-term (Beutel et al., 2010), pragmatic (Gawrysiak et al., 2013), and multimodal (de Greck et al., 2013) psychodynamic psychotherapies. This fact is a relevant methodological limitation for the comparison of results.

Each study used a different method in selecting stimuli (or recall trauma): OPD (Fischmann et al., 2013; Wiswede et al., 2014), emotional linguistic go/nogo task (Goldstein et al., 2007; Beutel et al., 2010), Adult Attachment

Projective Picture System (AAP) (George and West, 2001; Buchheim et al., 2012), alternating passages of preferred and neutral music (Osuch et al., 2009; Gawrysiak et al., 2013), and emotional faces stimuli taken from the 'Japanese and Caucasian Facial Expressions of Emotion' battery (JACFEE) and the 'Japanese and Caucasian Neutral Faces' battery (JACNeuF) (Matsumoto and Ekman, 1988).

Finally, some important limitations of the studies investigated in this systematic review should be stressed. The number of participants was too small. In fact, the total number of patients and controls were, respectively, 76 and 85. The studies used different psychotherapy treatments and different treatment durations. Two studies used a psychoanalytic treatment (Buchheim et al., 2013; Fischmann et al., 2013), and other studies used multimodal (de Greck et al., 2013), pragmatic (Gawrysiak et al., 2013), short- or long-term psychodynamic therapies (Beutel et al., 2010). Different methods to extract individualized and personally relevant stimuli were used. Some studies used OPD (Buchheim et al., 2013; Fischmann et al., 2013; Wiswede et al., 2014), others used AAP (Buchheim et al., 2012), JACFEE (de Greck et al., 2013), JACNeuF (de Greck et al., 2013), emotional linguistic go/nogo task (Beutel et al., 2010), or alternating passages of preferred and neutral music (Gawrysiak et al., 2013). Only two studies did not compare patients and healthy controls (Buchheim et al., 2013; Gawrysiak et al., 2013). Participant studies had different diagnoses: dysthymia with narcissistic traits, depressive, somatoform, or panic disorders. These conditions caused great difficulties in comparing results of the studies.

Finally, it is important to underline if patients enrolled in the studies were free of psychotropic medication or not at the time of the study. Some studies excluded patients with a current psychotropic treatment (Buchheim et al., 2012; Fischmann et al., 2013; Wiswede et al., 2014), other studies did not report if patients were taking any psychotropic drugs (Buchheim et al., 2013; Gawrysiak et al., 2013), and some authors enrolled patients under a psychopharmacological treatment (Beutel et al., 2010; de Greck et al., 2013).

Conclusions

In recent years, a connection between psychoanalysis and neuroscience has been sought. The meeting point between these two branches is neuropsychanalysis. Until now, a small number of studies that compared functional magnetic resonance imaging results before and after psychoanalytic and psychodynamic therapies have

been conducted. The common results from the studies investigated in this systematic review are represented by a reduction, a modulation, or a normalization of the activation patterns found before and after treatments. The studies investigated, show some limitations: the number of participants was too small, two studies did not compare patients and healthy controls, psychotherapy types and the duration of the treatments were different, and different methods to extract individualized and personally relevant stimuli were used. For these reasons, further methodological correct studies are needed. Researchers should keep in mind that in any scientific research that has to do with people, neuroscience and a scientific method cannot avoid subjective interpretation.

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